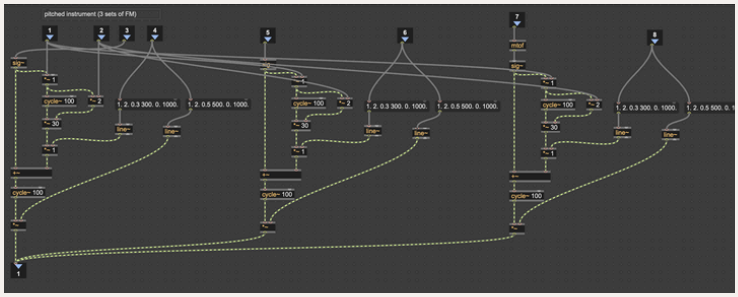
Amber Shi

2026

SIGNAL MOVEMENT RESONANCE

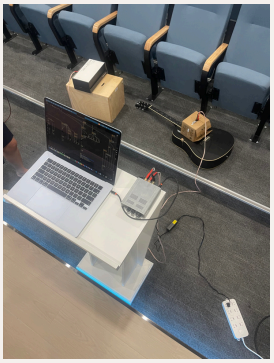
Introduction

A percussion instrument makes sound when an object, such as a drumstick, strikes a resonating surface. My initial goal for this project was to recreate that mechanism without drumsticks, using digital impulses to activate physical resonators and produce acoustic sound.



About the Resonators

All the resonators were made from old, recycled materials: cardboard boxes, a secondhand acoustic guitar body bought for about $5, and a tambourine. I attached speakers to each object and used a power amplifier to drive them.





Generative Composition & Audio Synthesis

Because the resonators changed the raw digital audio through their natural acoustic properties, I had to design the impulse sounds to complement those effects.

I synthesized four sounds in Max/MSP: three percussive impulses built from noise, and one melodic instrument created with frequency modulation (FM).

Motion Controls

The base sound consists of the percussion (kick, snare, and hi-hat at half tempo) and the lower voices of the FM synthesizer. Using MediaPipe in Max/MSP, I tracked two types of motion that control the hi-hat's speed and the FM synthesizer's upper voice.

Amount of Motion: I developed an algorithm that measures how much the image changes from frame to frame by counting pixel changes. When there is little motion, the hi-hat stays at half tempo and the upper voice, which carries the main melody, is muted. When there is more motion, the hi-hat switches to double tempo and the main melody comes in.

Height of Left Wrist: I developed a second algorithm that tracks the height of my left wrist. I divided its range into three zones (low, middle, and high), and each zone triggers a different set of pitches for the main melody.

